

Cloud Computing Reference Architecture

ITU-T Y.3502 | ISO/IEC 17789

Scope and concept

- Specify a cloud computing reference architecture (CCRA), including the **cloud computing roles, cloud computing activities**, and the **cloud computing functional components** and their relationships.
- The CCRA provides an architectural framework that is effective for describing the **cloud computing roles, sub-roles, cloud computing activities**, cross-cutting aspects, as well as the functional architecture and **functional components** of **cloud computing**.
- The CCRA focuses on the requirements of "what" **cloud services** provide and not on "how to" design cloud-based solutions and implementations.
- The CCRA does not represent the system architecture of a specific **cloud computing** system, although it could put constraints on a specific system.
- CCRA can serve as a fundamental reference point for **cloud computing** standardization and which provides an overall framework for the basic concepts and principles of a **cloud computing** system.

Goals and objective (1/2)

The CCRA serves the following **goals**:

- to describe the community of stakeholders for cloud computing;
- to describe the fundamental characteristics of cloud computing systems;
- to specify basic cloud computing activities and functional components, and describe their relationships to each other and to the environment;
- to identify principles guiding the design and evolution of the CCRA.

The CCRA supports the following important standardization **objectives**:

- to enable the production of a coherent set of international standards for cloud computing;
- to provide a technology-neutral reference point for defining standards for cloud computing;
- to encourage openness and transparency in the identification of cloud computing benefits and risks.

Goals and objective (2/2)

The CCRA is also intended to:

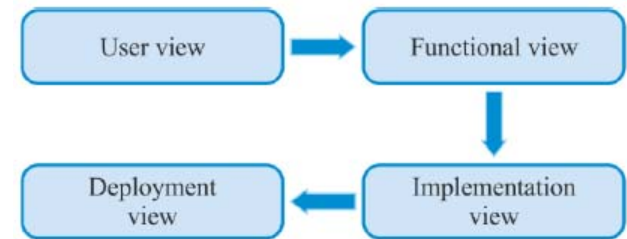
- facilitate the understanding of the operational intricacies of cloud computing;
- illustrate and provide understanding of various cloud services and their provisioning and use;
- provide a technical reference to enable the international community to understand, discuss, categorize and compare cloud services;
- be a tool for describing, discussing, and for developing a system-specific architecture using a common framework of reference;
- facilitate the analysis of candidate standards in areas including security, interoperability, portability, reversibility, reliability and service management, and support analysis of reference implementations.

CCRA architectural views

Cloud computing systems can be described using a viewpoint approach.

Four distinct viewpoints are used in the CCRA:

- **User view:** the system context, the parties, the roles, the sub-roles and the cloud computing activities
- **Functional view:** the functions necessary for the support of cloud computing activities
- **Implementation view:** the functions necessary for the implementation of a cloud service within service parts and/or infrastructure parts
- **Deployment view:** how the functions of a cloud service are technically implemented within already existing infrastructure elements or within new elements to be introduced in this infrastructure

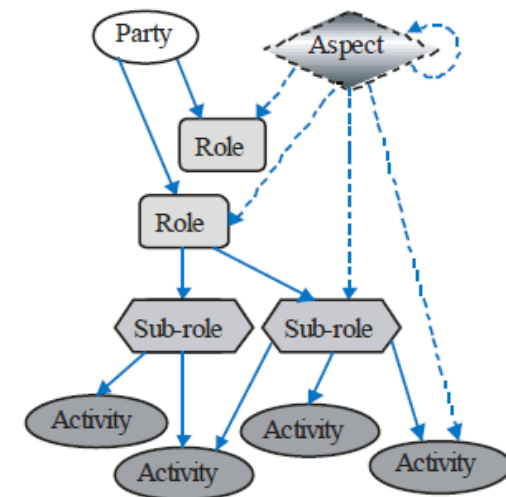


NOTE – The implementation and deployment views are related to technology and vendor-specific cloud computing implementations and actual deployments, and are therefore out of the scope of this Standard.

User view of cloud computing(1/2)

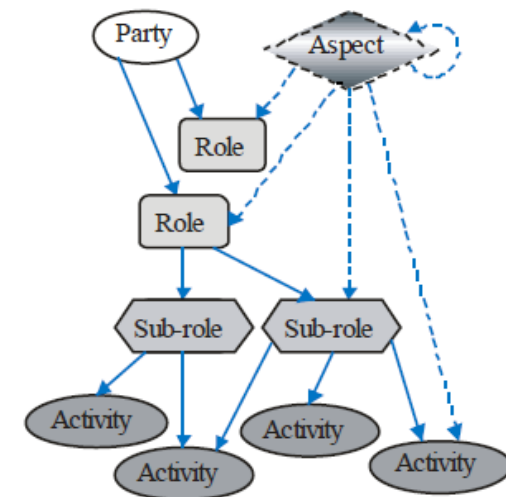
The **user view** addresses the following cloud computing concepts:

- Cloud computing activities:
 - A cloud computing activity is defined as a specified pursuit or set of tasks.
 - Cloud computing activities need to have a purpose and deliver one or more outcomes.
 - Activities in a cloud computing system are conducted using functional components
- Roles and sub-roles:
 - A role is a set of cloud computing activities that serve a common purpose.
 - A sub-role is a subset of the cloud computing activities for a given role.
 - Different sub-roles can share the cloud computing activities associated with a given role.
 - In the CCRA, three roles have been defined; *cloud service customer (CSC)*, *cloud service provider (CSP)* and *cloud service partner (CSN)*.
- Parties:
 - A party is a natural person or legal person, whether or not incorporated, or a group of either.
 - Parties in a cloud computing system are its stakeholders.
 - A party can assume more than one role at any given point in time and can engage in a specific subset of activities of that role.



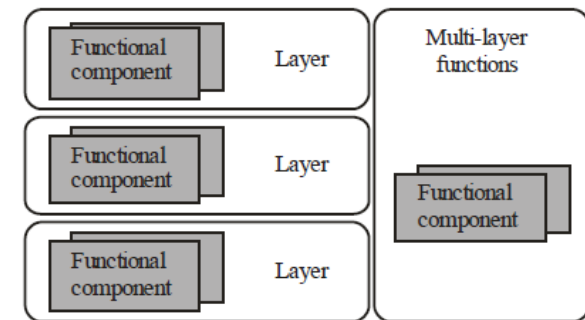
User view of cloud computing (2/2)

- Cloud services:
 - Cloud services are the essential elements of cloud computing.
 - Cloud services can be described in terms of the cloud capabilities types which they offer, based on the resources provided by the cloud service; e.g., *application capabilities type*, *platform capabilities type* and *infrastructure capabilities type*.
 - Cloud services are also grouped into categories, where each category is a group of cloud services that possess a common set of qualities; e.g., *IaaS*, *PaaS*, *SaaS* and, *NaaS*.
 - The services in these categories can include capabilities from one or more of the cloud capabilities types.
- Cloud deployment models:
 - Cloud deployment models are a way in which cloud computing can be organized based on the control and sharing of physical or virtual resources.
 - The cloud deployment models include; *public*, *private*, *community* and *hybrid*.
- Cross-cutting aspects:
 - Cross-cutting aspects are behaviors or capabilities which need to be coordinated across roles and implemented consistently in a cloud computing system.
 - Cross-cutting aspects can be shared and can impact multiple roles, cloud computing activities and functional components.
 - Cross-cutting aspects apply to multiple individual roles or functional components.
 - e. g., Security.



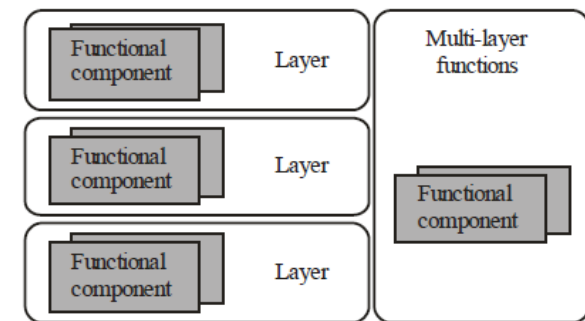
Functional view of cloud computing (1/2)

- The functional view is a technology-neutral view of the functions necessary to form a cloud computing system.
- The functional view describes the distribution of functions necessary for the support of cloud computing activities.
- The functional architecture also defines the dependencies between functions.
- The functional view addresses the following cloud computing concepts:
 - Functional components.
 - functional layers.
 - multi-layer functions.

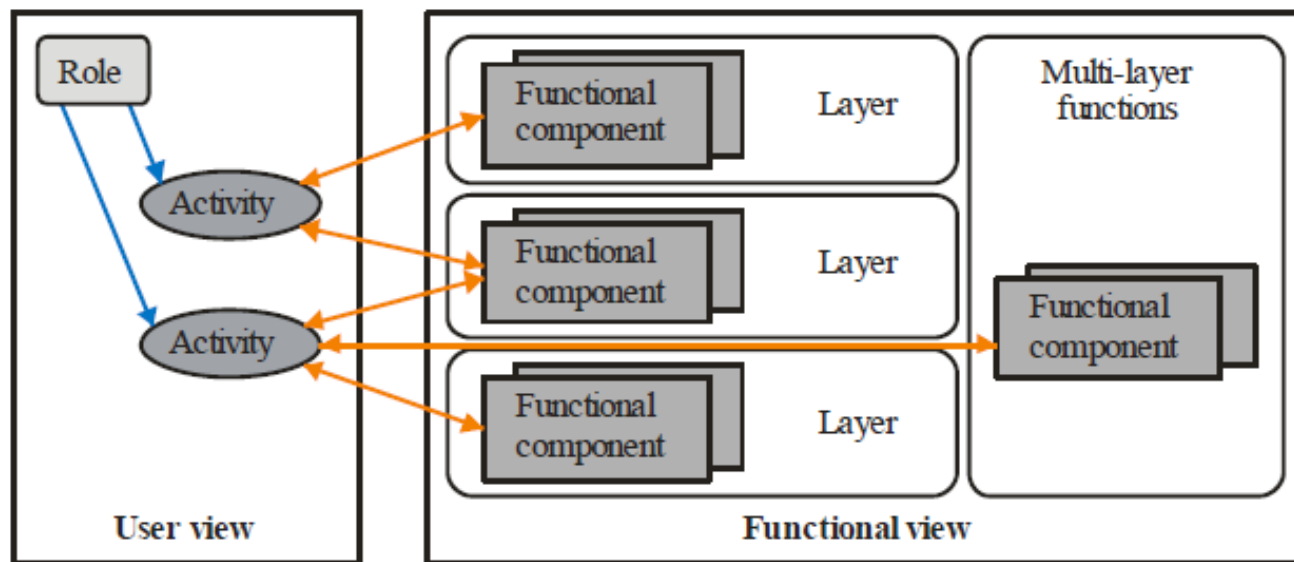


Functional view of cloud computing (2/2)

- Functional components:
 - A functional component is a functional building block needed to engage in an activity, backed by an implementation.
 - The capabilities of a cloud computing system are fully defined by the set of implemented functional components.
- Functional layers:
 - A layer is a set of functional components that provide similar capabilities or serve a common purpose.
 - There are four distinct layers defined in the CCRA:
 - **User layer**, which includes functional components that support the cloud computing activities of cloud service customers and cloud service partners.
 - **Access layer**, which includes functional components that facilitate function distribution and interconnection;
 - **Service layer**, which includes functional components that provide the cloud services themselves plus related administration and business capabilities, and the orchestration capabilities necessary to realize them.
 - **Resource layer**, which includes the functional components that represent the resources needed to implement the cloud computing system.
- multi-layer functions.
 - The multi-layer functions include functional components that provide capabilities that are used across multiple functional layers.



Relationship between the user view and the functional view



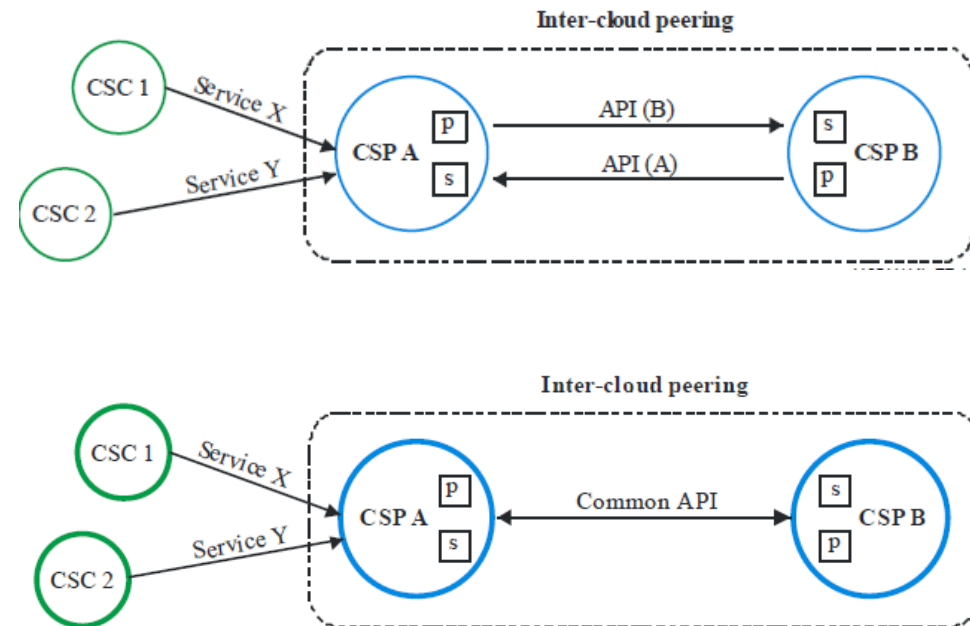
The user view provides the set of cloud computing activities that are represented within the functional view (and realized using the technologies of the implementation view).

Framework of inter-cloud computing

ITU-T Y.3511

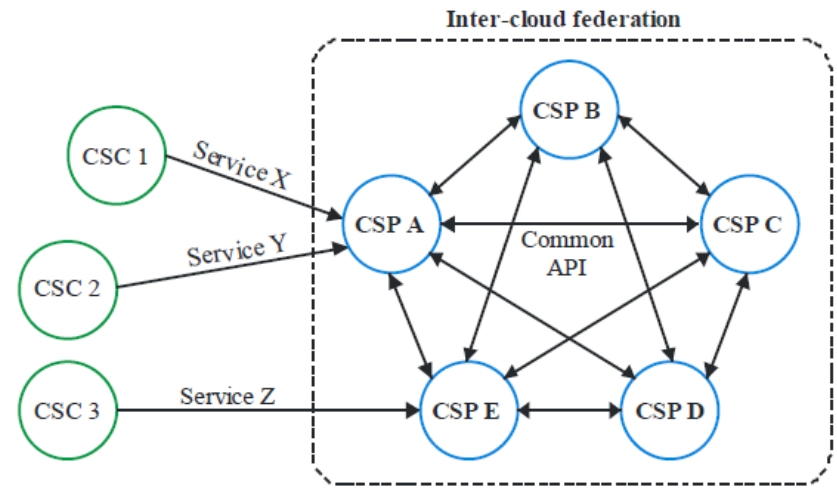
Patterns of inter-cloud: Inter-cloud peering

- In inter-cloud peering, two CSPs interwork directly with each other in order to use the services provided by the peer CSP.
- Inter-cloud peering does not necessarily imply reciprocal relationships in terms of service use and service providing between the two CSPs.
- Inter-cloud peering is a fundamental pattern, which may exist on its own or can be used in patterns; *inter-cloud federation* or *inter-cloud intermediary*.
- In inter-cloud peering, each CSP exposes its own API for cloud interworking and the CSPs interwork with each other directly by using the other CSP's API. On the other hand, a common API can be used between CSPs.



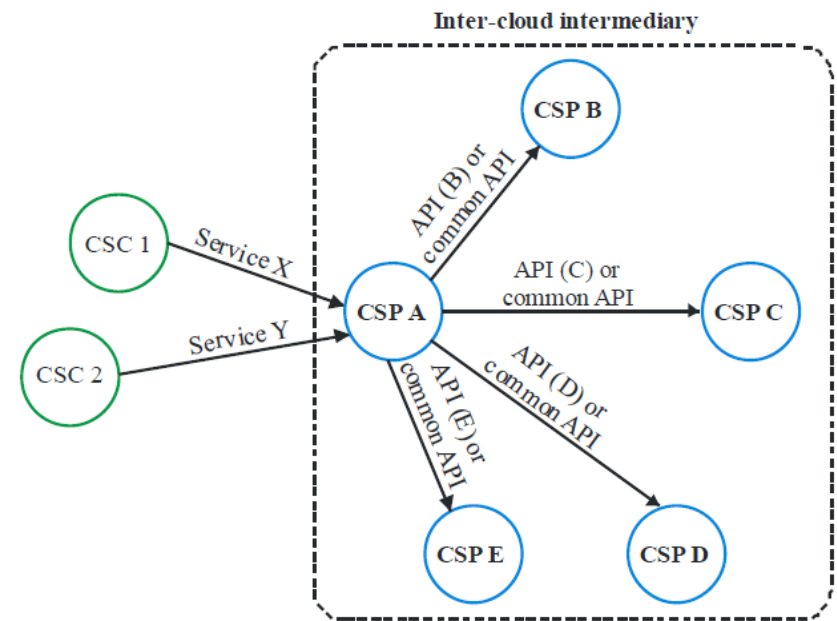
Patterns of inter-cloud: Inter-cloud federation

- Inter-cloud federation involves using the cloud services within a group of peer CSPs who mutually combine their service capabilities in order to provide the set of cloud services required by cloud service customers (CSCs).
- The multiple CSPs, which form the inter-cloud federation, establish and share the common agreement which may range from service-related policies, service level agreements (SLAs) and procedures which are relevant to the service offering and resource handling.
- A common API for cloud interworking is defined in the inter-cloud federation. Each CSP interworks with the other CSPs in the inter-cloud federation through this common API.



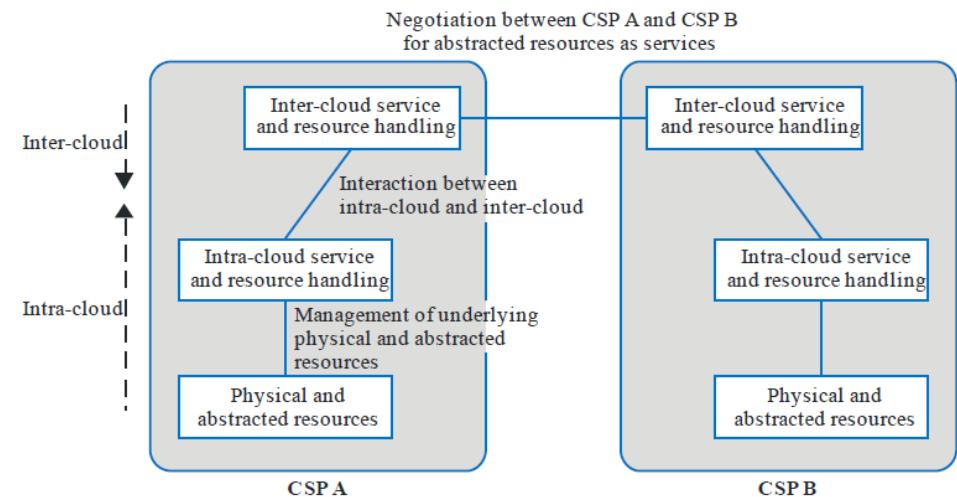
Patterns of inter-cloud: Inter-cloud intermediary

- In the inter-cloud intermediary pattern, the CSP interworks with one or more peer CSPs and provides intermediation, aggregation and arbitrage of services provided by these CSPs.
- Service intermediation consists in conditioning or enhancing the cloud service of a peer CSP.
- Service aggregation relates to the composition of a set of services provided by the peer CSPs.
- Service arbitrage is about selecting one service offering from a group of services offered by the peer CSPs.



Relationship between intra-cloud and inter-cloud handling of resources

- For collaboration among CSPs, two types of resources can be distinguished:
 - One includes underlying physical resources of a cloud infrastructure which are controlled and managed by the CSP who owns these resources.
 - The other type includes resources which are abstracted from the underlying physical resources and are offered as services to CSPs.
- Intra-cloud service and resource handling allows a CSP (A or B) to manage its own resources including underlying physical resources and abstracted resources.
- Inter-cloud service and resource handling allows a CSP to negotiate for the use of peer CSPs' abstracted resources provided as cloud services.



Functional requirements for inter-cloud

- **SLA and policy negotiation**: the SLA and policy negotiation capability deals with the matching made by a primary CSP between SLA requirements of a CSC and the SLAs of secondary CSPs involved in the considered inter-cloud pattern.
- **Resource monitoring**: deals with the monitoring by the primary CSP of the resources of the secondary CSPs and the status attributes of these resources (e.g., usage amount, performance and quality aspects).
- **Resource performance estimation and selection**: deals with the selection of resources from the candidate resources that have already been reserved in peer CSPs.
- **Resource discovery and reservation**: deals with search, discovery and reservation of available resources in the peer CSPs.
- **Resource set-up and activation**: deals with the set up and activation of reserved resources in the peer CSPs.
- **Cloud services switchover and switchback**: deals with the switchover of the CSC's end-user access to cloud services from the primary CSP to a peer CSP to which the services may be provided in order to cope with degradation in service performance or a serious problem. It also deals with the switchback to the primary CSP when it is able to provide the services again.
- **Resource release**: deals with the release by a CSP of the peer CSPs' (reserved and/or used) resources after determining that these resources are no longer needed.
- **CSC information exchange**: deals with exchanging CSC profiles and associated information between a primary CSP and the secondary CSPs.
- **Primary CSP role delegation**: deals with transferring the primary CSP role to one of the secondary CSPs.
- **Inter-cloud service handling**: deals with the primary CSP offering cloud services to its CSCs based on the handling of services provided by the secondary CSPs.